

RAFFLES INSTITUTION
2021 Year 6 Preliminary Examination

Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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BIOLOGY

Paper 2 Structured Questions

9744/02

22nd September 2021
2 hours

Candidates answer on Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name in the spaces at the top of this page.

Write in a dark blue or black pen.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

1	/ 10
2	/ 9
3	/ 11
4	/ 12
5	/ 14
6	/ 11
7	/ 14
8	/ 9
9	/ 10
Total	/ 100

This document consists of **28** printed pages.



Raffles Institution
Internal Examination

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 represents the molecular structure of a transmembrane receptor on the cell surface membrane. It is made up of 485 amino acid residues.

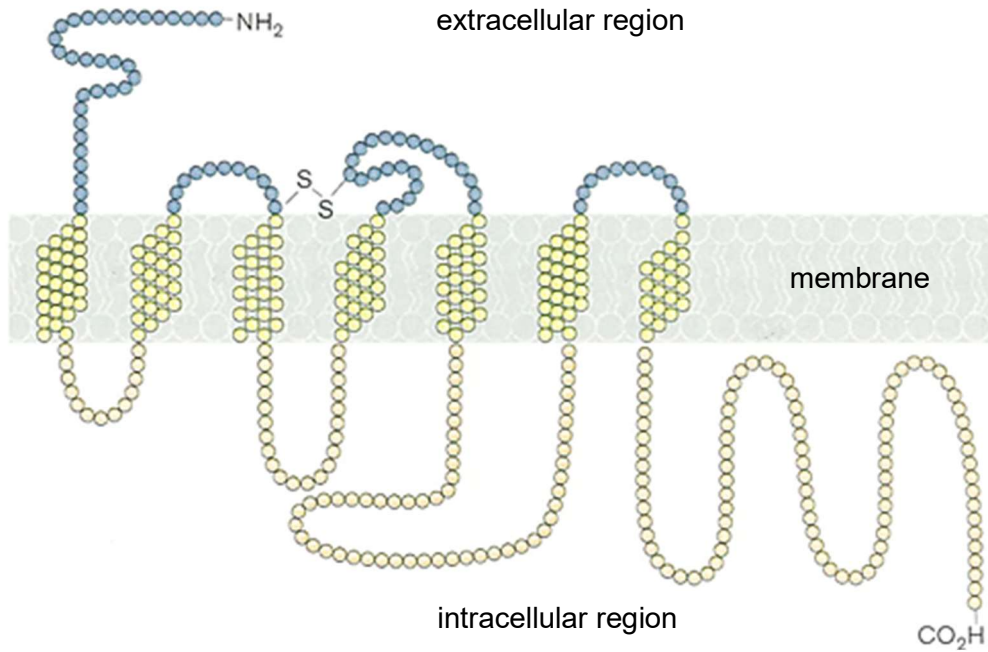


Fig.1.1

- (a) Name the type of receptor.

.....[1]

- (b) Describe the various levels of organisation in the structure of this receptor.

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.....[4]

- (c) A group of cell surface membrane proteins which transport sugars out of a cell have been identified and called SWEETs. They can be found in both plant and animal cells.

Each SWEET is a protein with seven coiled regions which together make a pore across a membrane bilayer shown in Fig. 1.2.

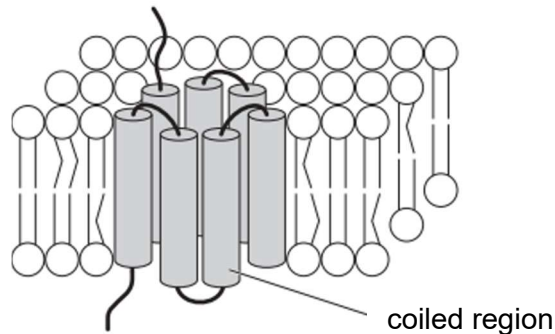


Fig. 1.2

- (i) Compare the visible structures of the transmembrane proteins shown in Fig. 1.1 and Fig. 1.2.

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..... [2]

- (ii) How do the transmembrane proteins in Fig. 1.1 and 1.2 differ in their roles?

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..... [2]

- (iii) Name a type of cell in an animal that has both the transmembrane proteins shown in Fig. 1.1 and 1.2.

..... [1]

[Total: 10]

- 2 Fig. 2.1 shows part of a macrophage seen under an electron microscope.

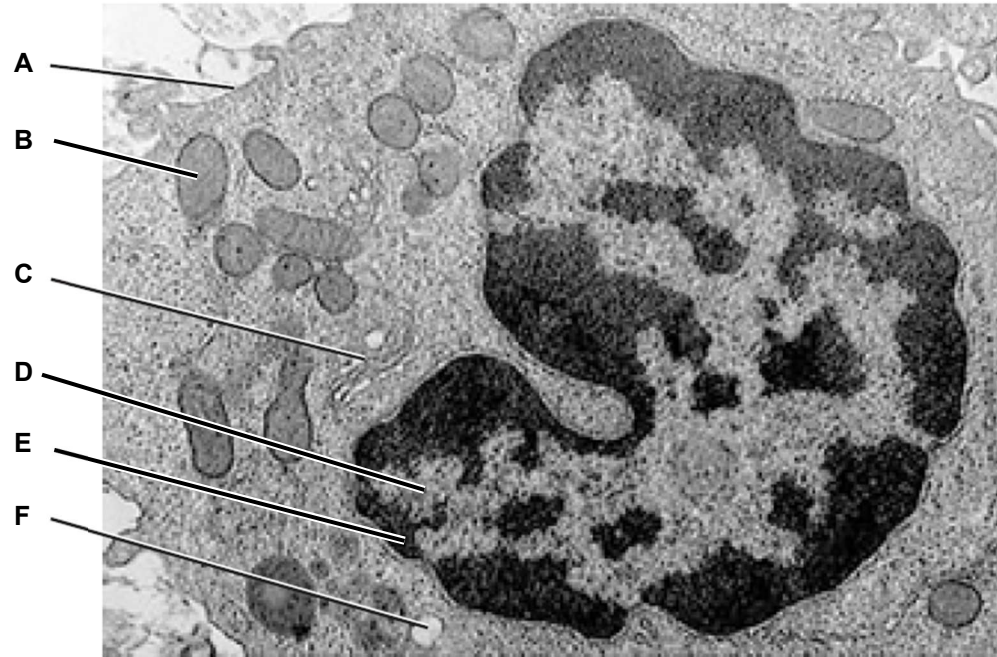


Fig. 2.1

- (a) (i) The table below gives information about the cell structures labelled in Fig. 2.1.

Complete the table by filling in the empty boxes.

structure labelled	name of structure	description of structure
B	mitochondrion	
		Loosely coiled DNA that is transcriptionally active.

[2]

- (ii) Explain the importance of structure A in a macrophage during phagocytosis.

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.....[2]

- (b) An experiment was carried out to study the synthesis and movement of proteins within macrophages.

Fig. 2.2 shows the method used and Table 2.1 shows the results of this experiment.

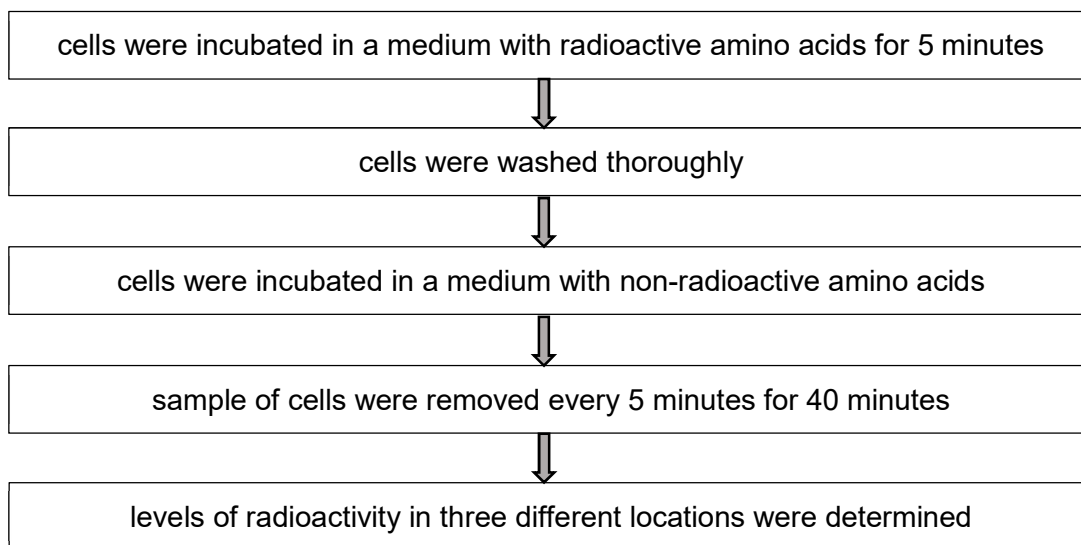


Fig. 2.2

Table 2.1

time / minutes	level of radioactivity / arbitrary units		
	ribosomes	cisternal space of rough endoplasmic reticulum	Golgi apparatus
5	60	0	0
10	80	40	0
15	20	50	0
20	15	50	10
25	10	20	30
30	5	20	60
35	5	10	40
40	5	5	20
total radioactivity	200	195	160

- (i) Describe and explain the changes in the level of radioactivity in the ribosomes.

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.....[3]

- (ii) Suggest why the total radioactivity in the Golgi apparatus was lower than the total radioactivity in the ribosomes.

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.....[2]

[Total: 9]

- 3 Fig 3.1 shows the intermediates and key stages of the Calvin cycle.

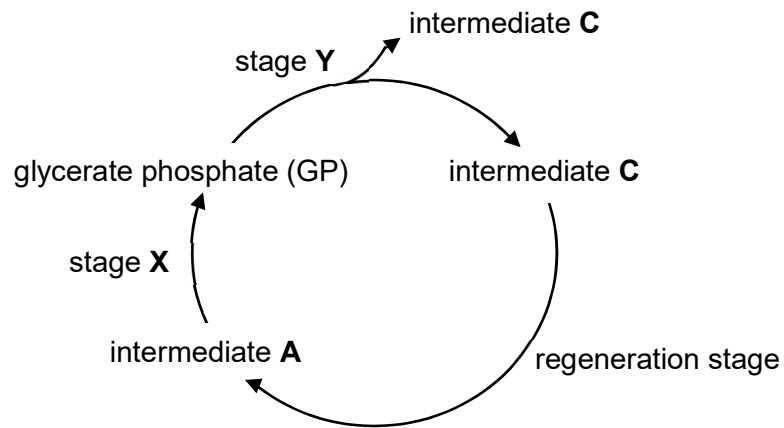


Fig 3.1

- (a) (i) Identify intermediates **A** and **C**.

intermediate **A**

intermediate **C**[1]

- (ii) Name the molecule(s) required to convert GP to intermediate **C**.

.....[1]

Fig 3.2 shows the results of an experiment. Unicellular green algal culture was grown in ample light under normal CO_2 conditions. At the time indicated in Fig. 3.2, radioactive $^{14}\text{CO}_2$ was added to the algal culture. The levels of radioactivity of intermediate A and GP were measured throughout the experiment.

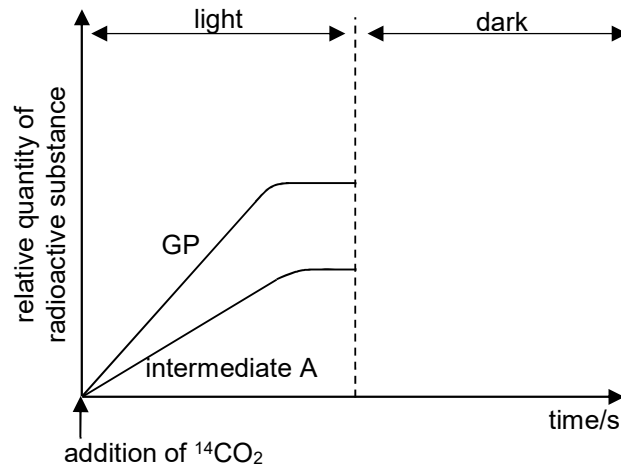


Fig. 3.2

- (b) With reference to Fig 3.2, explain the increase in the quantity of radioactive GP.

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.....[3]

- (c) In the middle of the experiment, the algal culture was placed in the dark.

On Fig 3.2, extend and label the lines to predict the changes in the quantity of

- (i) GP
- (ii) intermediate A

when placed in the dark.

[2]

(d) Explain your predicted trends given in part (c).

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.....[4]

[Total: 11]

- 4 In natural environments, bacteria such as *Escherichia coli* (*E. coli*) predominantly live in communities termed biofilms, in which a self-produced extracellular matrix holds the cells together. Bacterial biofilms are found in virtually every ecosystem on Earth, from aquatic systems to terrestrial environments and the human body. Furthermore, biofilms are associated with persistent and severe infections because of their ability to colonise medical devices and implants. The biofilm structures offer the bacteria protection against hostile environments and, more worryingly, against antibiotic treatments.

The biofilm environment provides a high cell density and close cell-cell proximity that may facilitate gene transfer through bacterial conjugation.

(a) (i) Compare the processes of bacterial conjugation and transformation.

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.....[2]

(ii) Conjugation drives the evolution of bacteria.

Explain why this is so.

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.....[2]

- (b) During conjugation in *E. coli*, the donor cell transfers an F plasmid to the recipient cell.

Fig. 4.1 shows the structure of the *tra* operon found on the F plasmid.

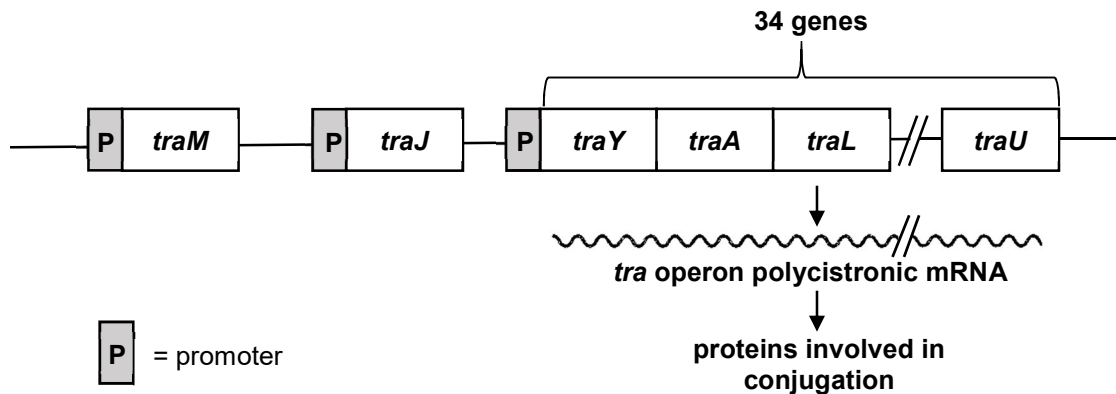


Fig. 4.1

- The majority of *tra* genes are transcribed under a single promoter as part of a large operon. There is no operator sequence in this operon.
- Regulation of the *tra* operon is controlled by the *traJ* gene which is located upstream of the operon.
- *traJ* encodes a protein that binds to the promoter of the *tra* operon to activate it.
- *traM* encodes a relaxosome protein which makes a site-specific DNA cut (nick) on one of the strands of the double-stranded plasmid at the origin of transfer.

- (i) Describe the role of the *tra* operon polycistronic mRNA.

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.....[2]

- (ii) Describe the process by which the relaxosome protein facilitates conjugation.

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.....[3]

Recent studies have found that the *tra* operon is repressed by bacterial host-nucleoid associated (H-NS) proteins which are encoded by the bacterial chromosome. Fig. 4.2 shows the location where the H-NS protein binds to on the operon.

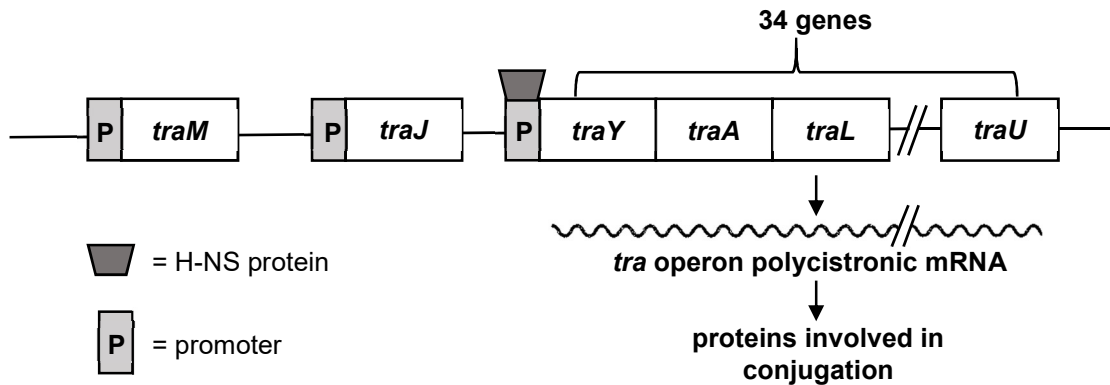


Fig 4.2

(iii) With reference to Fig. 4.2, explain how the H-NS protein represses the *tra* operon.

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.....[2]

(c) Another gene, *traT*, is located at a separate locus on the F plasmid. This gene is not required for sex pilus synthesis and DNA transfer. Instead, *traT* codes for a membrane protein.

When expressed in a bacterial cell, the TraT membrane protein prevents a sex pilus from another F⁺ cell from binding to it.

Suggest why it is advantageous for the F plasmid to contain the *traT* gene.

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.....[1]

[Total: 12]

- 5 In tomato plants, stem colour, leaf shape and leaf colour are controlled by three different genes that are found on the same chromosome. The allele for red stems, **R**, is dominant to the allele for green stems, **r**. The allele for tomato leaf shape, **T**, is dominant to the allele for potato leaf shape, **t** and the allele for dark green leaves, **D** is dominant to the allele for light green leaves, **d**. The two tomato plants described below were crossed together.

<i>Parent 1</i>	<i>Parent 2</i>
red stem	green stem
tomato leaf shape	potato leaf shape
dark green leaves	light green leaves

- (a) The parent plants were both pure-breeding.

State the genotype and phenotype of the F_1 plants.

genotype:

phenotype:[1]

- (b) The F_1 plants were then selfed to produce an F_2 generation of plants. Since the genes are linked, many of the offspring in the F_2 generation were expected to show mostly the two original parental phenotypes.

Assuming there is no crossing over between the linked genes,

- (i) state the genotypes of the F_2 plants showing the same phenotypic characteristics as parent 1;

.....[1]

- (ii) give the expected genotypic ratio of all the F_2 offspring.

.....[1]

In fact, there were eight different phenotypes among the F_2 plants. This can be explained by crossing over having taken place.

Fig. 5.1. represents the behaviour of one pair of chromosomes during meiosis.

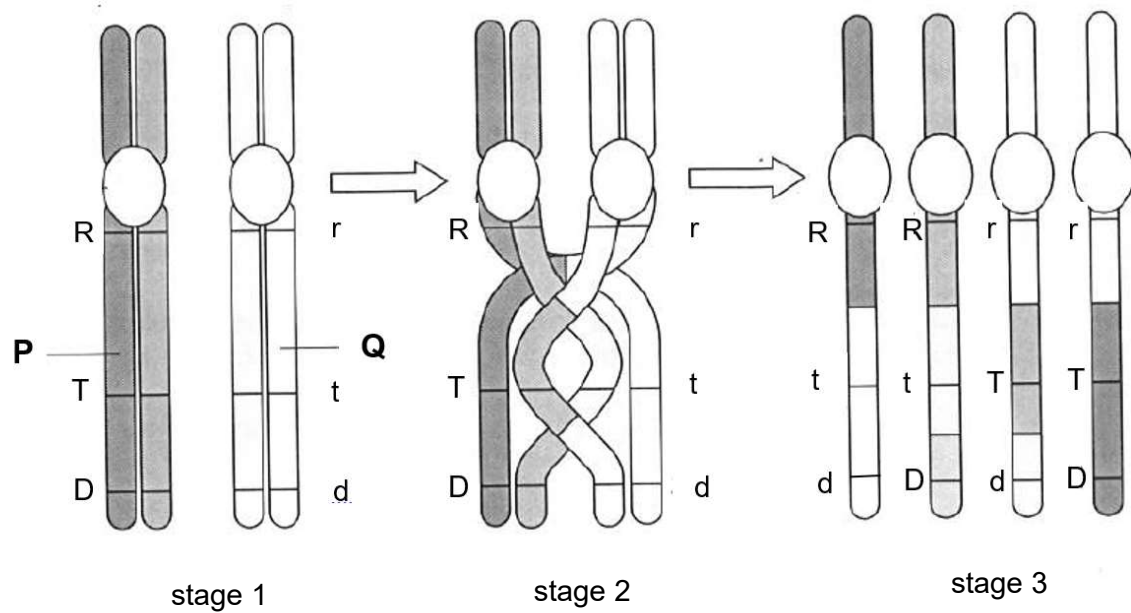


Fig. 5.1

(c) With reference to Fig. 5.1, compare structure **P** with structure **Q**.

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.....[3]

(d) Describe the events that occurred in stage 2 in Fig. 5.1.

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.....[2]

- (e) A farmer collected two tomato plants, T1 and T2, which were produced from crosses between different pure-breeding plants in his garden. Both have dark green leaves and red stems. He conducted a test cross on each plant. Table 5.1 shows the results of these crosses.

Table 5.1

phenotype	observed numbers	
	plant T1	plant T2
dark green leaves and green stems	122	2
light green leaves and green stems	6	82
dark green leaves and red stems	4	85
light green leaves and red stems	124	4

- (i) Draw a genetic diagram to explain the results of the test cross for plant T2.

[3]

- (ii) Account for the different test cross results obtained from plants T1 and T2.

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.....[3]

[Total: 14]

- 6 Telomere length is maintained in most eukaryotic cells by telomerase. The core components of this ribonucleoprotein (RNP) complex include a protein catalytic subunit (TERT), and an RNA subunit that contains a short template sequence essential for the synthesis of telomeric repeats.

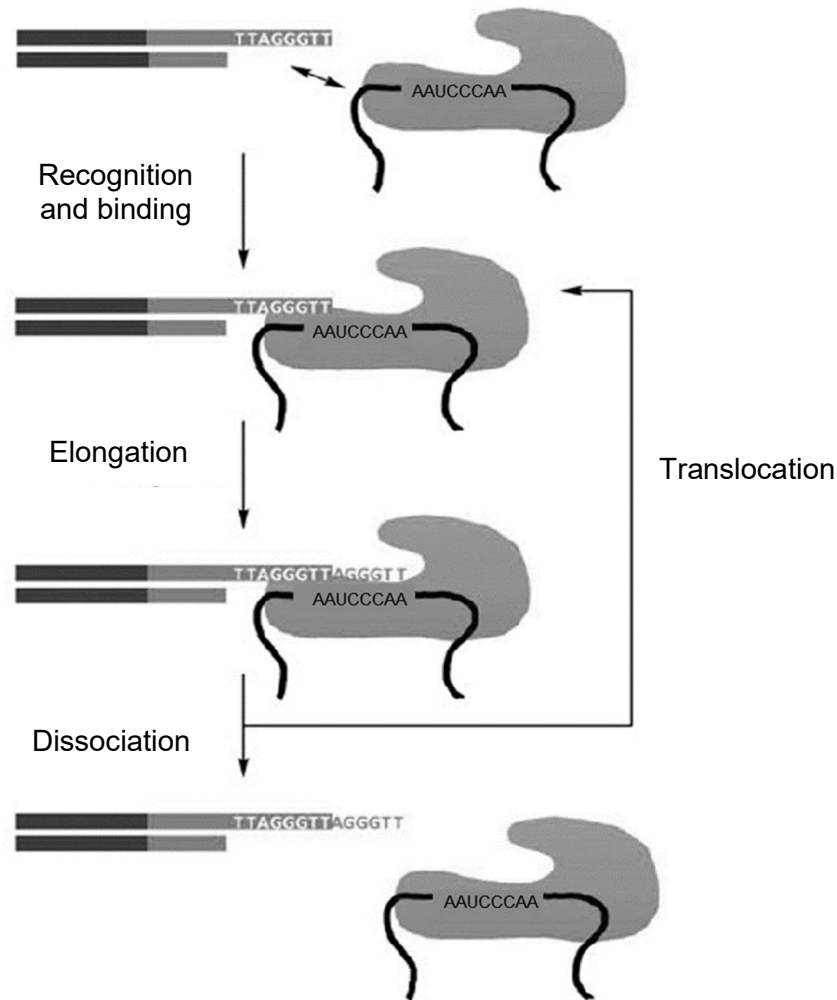


Fig. 6.1

- (a) (i) With reference to Fig. 6.1, explain the role of the TERT protein in telomerase.

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.....[3]

- (ii) Telomerase activity is observed to be high in stem cells but downregulated in differentiated cells.

Explain why high telomerase activity is needed in stem cells but not in differentiated cells.

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.....[3]

As with stem cells, cancer cells are also known to exhibit high levels of telomerase activity.

The high levels of telomerase activity in cancer cells may be contributed by mutations which involve the *TERT* locus, as shown in Fig. 6.2 below.

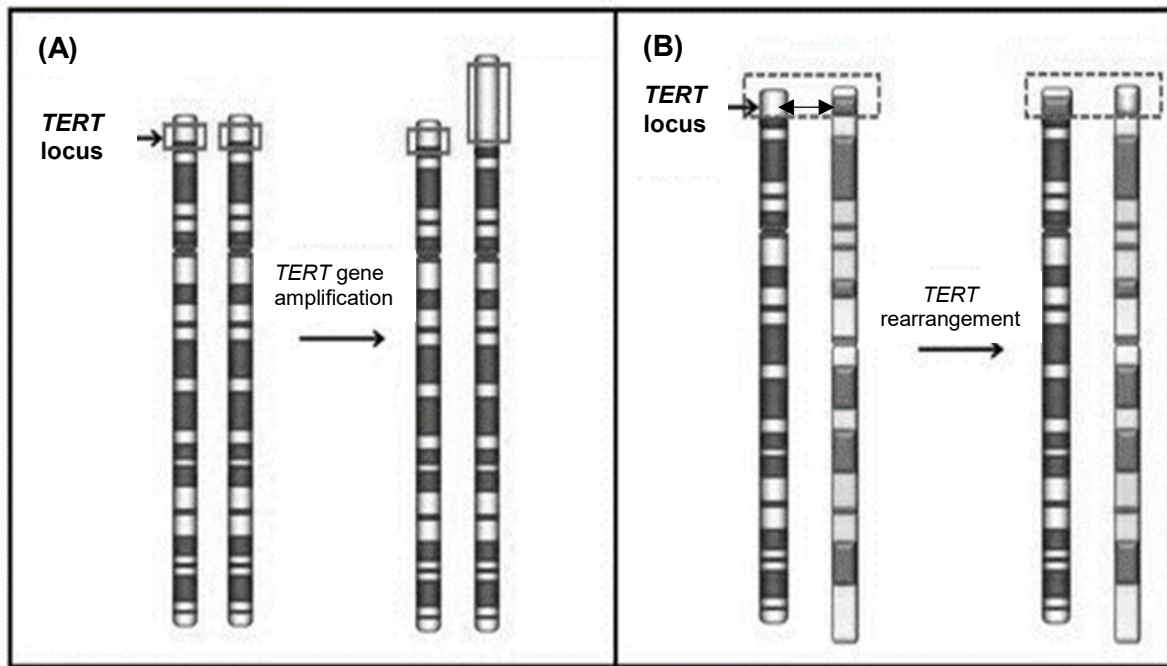


Fig. 6.2

- (b) Suggest how the mutations that involve the *TERT* locus shown in the Fig. 6.2(A) and Fig. 6.2(B) could result in the high levels of telomerase activity in cancerous cells.

- (i) Fig. 6.2(A):

.....
[1]

- (ii) Fig. 6.2(B):

.....
[1]

(c) In the laboratory, differentiated cells can be made into induced pluripotent stem cells (iPSCs). This is done by introducing transcription factors into the differentiated cells.

(i) Explain how transcription factors work in the production of iPSCs from differentiated adult cells.

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.....[2]

(ii) Explain how iPSCs overcome an ethical concern of using embryonic stem cells in therapy.

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.....[1]

[Total: 11]

7

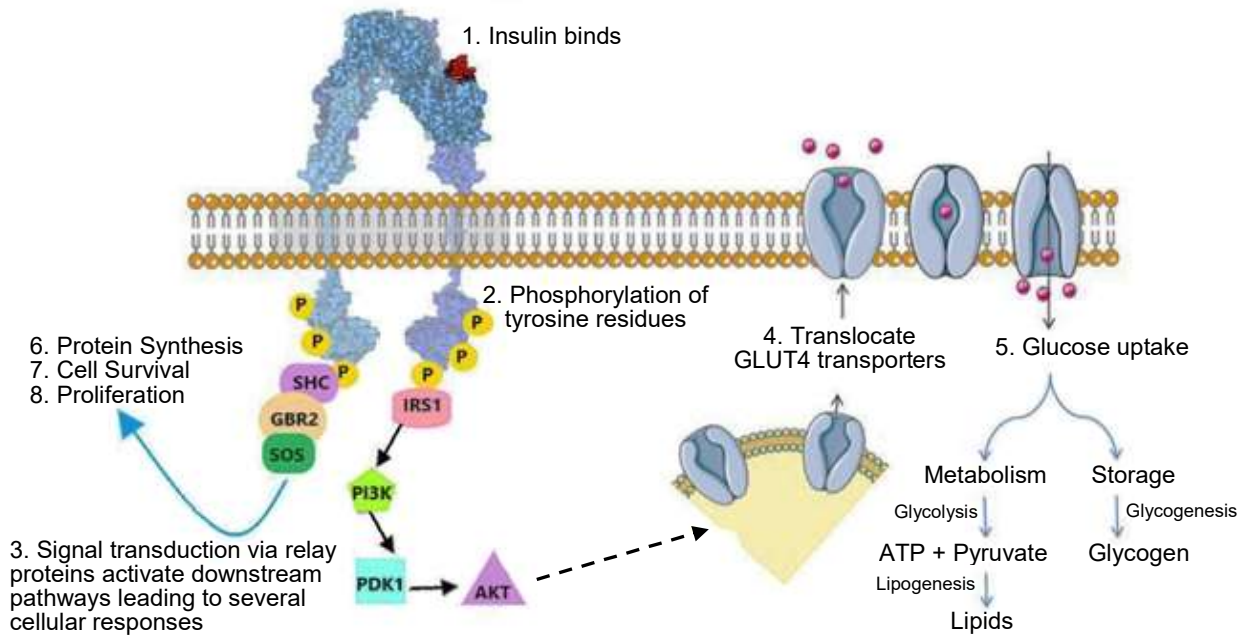


Fig. 7.1

With reference to Fig. 7.1,

- (a) (i)** describe how phosphorylation of tyrosine residues occur;

[3]

- (ii) suggest how activation of AKT can lead to an increase in glucose permeability of muscle cells.

.....[2]

- (b) AKT will also activate an enzyme, glycogen synthase, which will catalyse the conversion of glucose into glycogen.

Explain how the structure of glycogen relates to its role as a storage molecule.

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[2]

- (c) Another cellular response to insulin apart from glycogen synthesis is the increased rate of respiration of the muscle cells using glucose as a respiratory substrate.

Fig. 7.2 represents two stages of aerobic respiration that take place in a mitochondrion.

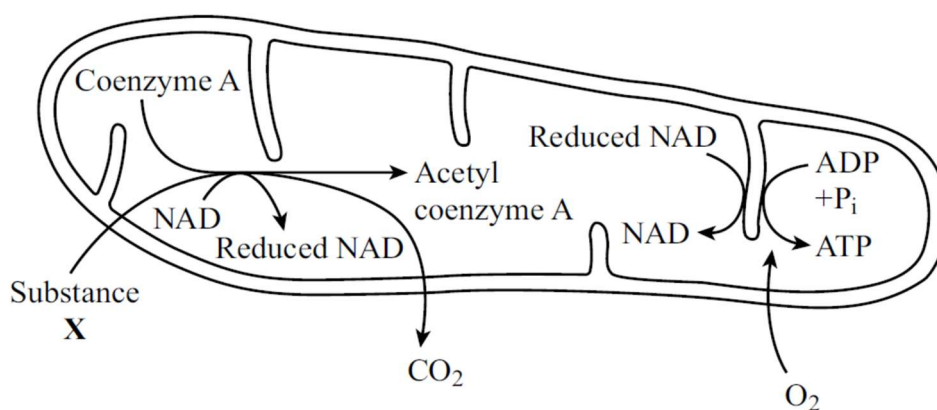


Fig. 7.2

- (i) Name substance X.

.....[1]

- (ii) Which stage of aerobic respiration takes place inside a mitochondrion but is not represented on the diagram?

.....[1]

- (iii) Explain why oxygen is needed for the production of ATP on the cristae of the mitochondrion.

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.....[3]

All human cell surface membranes and some internal membranes contain transmembrane glucose transporters (GLUTs).

There are at least 14 different GLUTs, each with differing functions. For example, GLUT1 ensures that sufficient glucose enters cells to supply their energy requirements.

- (d) Mature red blood cells do not contain mitochondria and have relatively low energy requirements. Despite this, the membranes of red blood cells contain more GLUT1 molecules than other cells with similar energy requirements.

Explain why the cell surface membranes of red blood cells require large numbers of GLUT1 molecules to supply their energy requirements.

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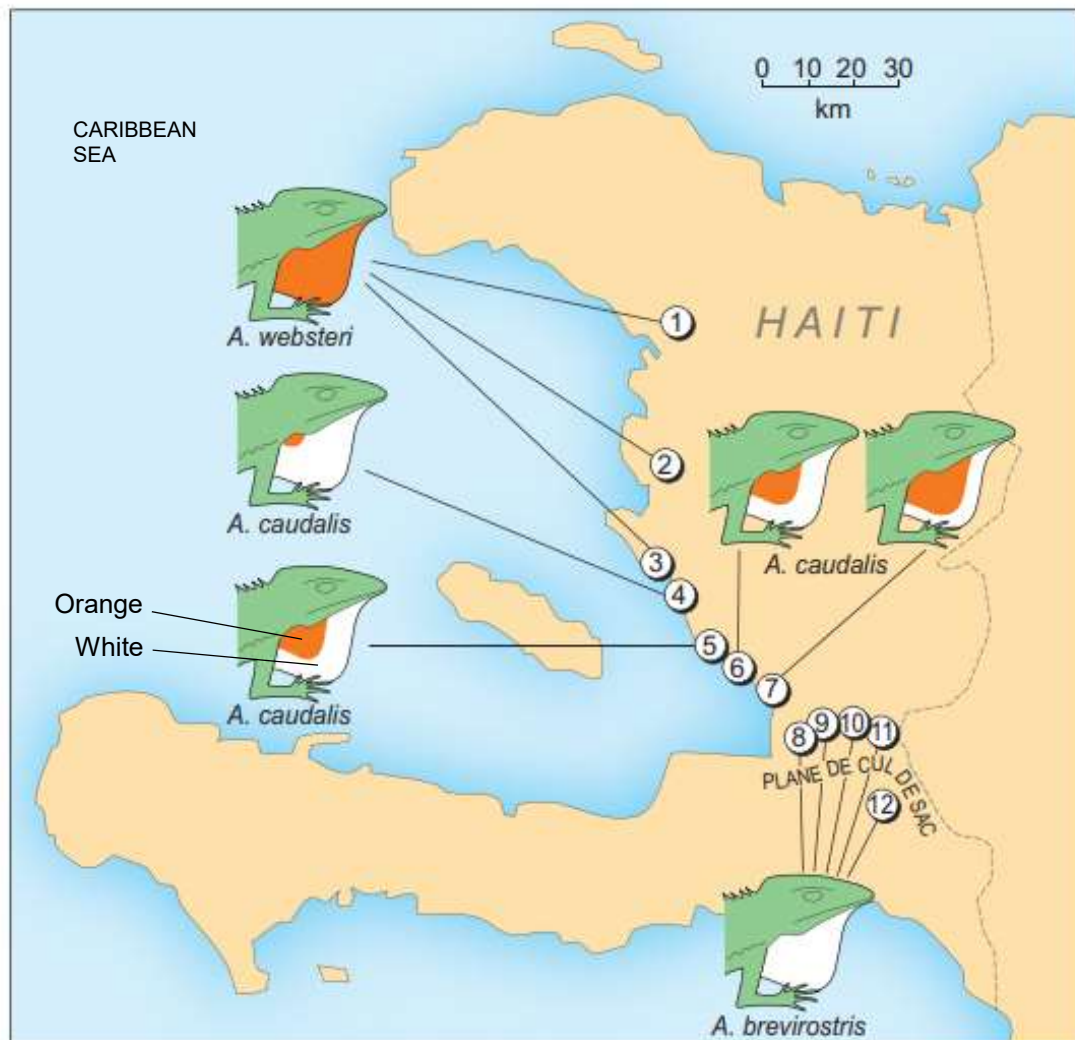
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.....[2]

[Total: 14]

- 8 In the Caribbean islands, there is a group of lizards called anoles. The males possess a larger retractable flap of skin under the throat, termed a dewlap, which is used in courtship, aggressive interactions and even as a warning when encountering predators.

The patterns of orange colouration on the dewlaps of 3 species of the anole lizard and the distribution of 12 population study sites in Haiti are shown below.



zone	population study sites	habitat
north	1, 2, 3	beach
central	4, 5, 6, 7	beach
south	8, 9, 10, 11, 12	forest

Fig. 8.1

(a) Define the term species.

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.....[2]

(b) The patterns of orange and white dewlap colouration for each anole species is different.

(i) Explain how the habitat in the south zone could bring about the dewlap colouration found in *A. brevirostris*.

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.....[4]

(ii) Breeding experiments in the lab have revealed that dewlap colour-pattern follows Mendel's laws of inheritance, in which orange (**D**) is dominant to white (**d**).

Although having a white colouration is disadvantageous at the beach habitats, the frequency of allele **d** in *A. websteri* has not reached zero even after many generations.

Explain why this is so.

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.....[2]

- (c) Suggest how anole lizards could have moved from Haiti to another island.

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[1]

[Total: 9]

- 9 The treatment for people with active tuberculosis (TB) lasts 6 months and involves a combination of antibiotics. This is usually very effective if the person has a susceptible (non-resistant) strain of *Mycobacterium tuberculosis*, the causative organism of TB.

Table 9.1 summarises one recommended treatment strategy that involves a combination of antibiotics.

Table 9.1

antibiotic	length of treatment	mode of action of antibiotic
rifampicin	6 months	enters bacterial cells and inhibits RNA polymerase activity
isoniazid	6 months	prevents the synthesis of cell wall components known as mycolic acids
ethambutol	first 2 months	prevents mycolic acids from being added to the cell wall
pyrazinamide	first 2 months	prevents the synthesis of fatty acids

- (a) Susceptible strains of *M. tuberculosis* will be killed using any one of the antibiotics listed in Table 9.1. However, combination treatment is preferred to reduce the prevalence of antibiotic resistance.

With reference to Table 9.1, explain how combination treatment for TB can help to reduce the prevalence of antibiotic resistance compared to single antibiotic treatment.

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[2]

- (b) Rifampicin binds tightly to an RNA polymerase molecule close to its active site. This affects the activity of the enzyme.

During the formation of mRNA, a number of events occur that involve the action of RNA polymerase.

Suggest how rifampicin can affect the activity of RNA polymerase.

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.....[3]

- (c) RNA polymerase is composed of 5 different polypeptides. Gene *rpoB* codes for one of these polypeptides known as the β -subunit.

One or more mutations in a specific region of *rpoB* result in strains of *M. tuberculosis* that are resistant to rifampicin. In these strains, mutations often occur in 2 DNA triplets in this region, at positions 526 and 531.

Table 9.2 summarises the results of an investigation into 7 rifampicin-resistant strains, **A** to **G**, that have amino acid changes for positions 526 and 531.

Table 9.2 includes:

- the change in the **mRNA codon** for position 526 or position 531
- the amino acid change that has occurred as a result of the mutation
- the minimum concentration of rifampicin required to inhibit growth of the bacterial strain (MIC)
- the number of **other** mutations occurring within the specific region of *rpoB*.

Table 9.2

Key: $\approx$ approximately $\geq$ greater than or equal to $\leq$ less than or equal to

strain	codon involved	mRNA codon change	amino acid change	MIC/ μgcm^{-3}	number of other mutations in the specific region
A	526	CAC $\rightarrow$ UAC	His $\rightarrow$ Tyr	≤ 50	0
B	526	CAC $\rightarrow$ AAC	His $\rightarrow$ Asn	≥ 100	1
C	526	CAC $\rightarrow$ CGC	His $\rightarrow$ Arg	$\approx 50-75$	2
D	526	CAC $\rightarrow$ CGC	His $\rightarrow$ Arg	≥ 100	3
E	526	CAC $\rightarrow$ CGC	His $\rightarrow$ Arg	≈ 50	3
F	526	CAC $\rightarrow$ UUC	His $\rightarrow$	≥ 100	3
	531	UCG $\rightarrow$ UUG	Ser $\rightarrow$ Leu		
G	526	CAC $\rightarrow$ UAC	His $\rightarrow$	≥ 100	3
	531	UCG $\rightarrow$ UUC	Ser $\rightarrow$ Phe		

- (i) Complete Table 9.2 to show the amino acid changes that have occurred in strains **F** and **G**. [1]
- (ii) With reference to Table 9.2, list the strains of *M. tuberculosis* that show the greatest resistance to rifampicin.

.....[1]

(c) Suggest reasons to explain why strains **C**, **D** and **E** show:

- resistance to rifampicin
- different levels of resistances to rifampicin.

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[Total:10]

End of Paper